# Cisco PE Node KR-PE-15 KR Puram Access Routing Protocol Flap Resolution SOP

**Node:** KR-PE-15 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_2224

## Description

This Standard Operating Procedure (SOP) addresses the failure domain of routing protocol flap on the Cisco PE node KR-PE-15 located at the KR Puram Access site. The routing protocol flap, indicated by alarm code ALARM_2224 with a severity of Warning, is a critical failure that requires immediate attention to prevent network instability and potential data loss. The root cause of this failure is typically attributed to issues within the BGP routing protocol, including route churn, OSPF adjacency loss, and ISIS LSP flooding. These issues can arise due to misconfiguration, hardware failures, or software bugs within the Cisco IOS-XE operating system. Understanding the underlying protocol mechanics, specifically the BGP route reflection and OSPF area border router roles, is essential in diagnosing and resolving this failure domain. Additionally, the impact of ISIS LSP flooding on the network topology and the potential for routing loops and black holes must be considered during the diagnostic and resolution process.

## Pre-requisite Checks

- Verify that the KR-PE-15 node is accessible via the console port or a secure remote access method.

- Check the redundant link status and confirm that the secondary link is operational.

- Gather relevant logs and configuration files from the KR-PE-15 node to aid in the diagnostic process.

- Notify the network operations center (NOC) and obtain approval for any necessary changes or maintenance.

- Ensure that the KR Puram Access site has a redundant power source and cooling system to prevent hardware failures due to power or environmental issues.

## Potential Root Causes

| Cause | Probability |
|---|---|
| BGP route reflection misconfiguration | High |
| OSPF adjacency loss due to network partitioning | Medium |
| ISIS LSP flooding due to hardware or software issues | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 1, local router ID is 10.10.10.1, Description: This command displays a summary of the BGP routing table, including the number of routes, neighbors, and the local router ID. This information can help identify any BGP route reflection issues.
```

```
Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: This command displays a list
of OSPF neighbors, including their ID, priority, state, dead time, and
interface information. This information can help identify any OSPF
adjacency loss issues.

Command: show isis database, Expected Output: ISIS LSP Database,
Description: This command displays the ISIS LSP database, including
information about the LSPs, their sequence numbers, and the
time-to-live (TTL). This information can help identify any ISIS LSP
flooding issues.
```

## Resolution Steps

Step 1: Step: Verify BGP route reflection configuration, Command: show running-config | include bgp, Description: This command displays the BGP configuration, including the route reflection configuration. Verify that the route reflection configuration is correct and consistent with the network design.

Step 2: Step: Reset BGP session, Command: clear ip bgp *, Description: This command resets the BGP session, including the route reflection and neighbor relationships. This can help resolve BGP route reflection issues.

Step 3: Step: Verify OSPF adjacency, Command: show ip ospf neighbor, Description: This command displays a list of OSPF neighbors, including their ID, priority, state, dead time, and interface information. Verify that the OSPF adjacency is established and stable.

Step 4: Step: Reset OSPF adjacency, Command: clear ip ospf neighbor *, Description: This command resets the OSPF adjacency, including the neighbor relationships and the OSPF database. This can help resolve OSPF adjacency loss issues.

Step 5: Step: Clear ISIS LSP database, Command: clear isis database *, Description: This command clears the ISIS LSP database, including the LSPs and their sequence numbers. This can help resolve ISIS LSP flooding issues.

## Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 1,
local router ID is 10.10.10.1, Description: This command displays a
summary of the BGP routing table, including the number of routes,
neighbors, and the local router ID. Verify that the BGP routing table
is stable and consistent with the network design.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: This command displays a list
of OSPF neighbors, including their ID, priority, state, dead time, and
interface information. Verify that the OSPF adjacency is established
and stable.

Command: show isis database, Expected Output: ISIS LSP Database,
Description: This command displays the ISIS LSP database, including
information about the LSPs, their sequence numbers, and the
time-to-live (TTL). Verify that the ISIS LSP database is stable and
consistent with the network design.
```

## Roll-back Procedures

- Step: Restore previous configuration, Command: copy running-config startup-config, Description: This command restores the previous configuration from the startup configuration file. This ensures that the network is returned to its previous stable state.

- Step: Reset BGP session, Command: clear ip bgp *, Description: This command resets the BGP session, including the route reflection and neighbor relationships. This can help resolve BGP route

reflection issues.

- Step: Reset OSPF adjacency, Command: clear ip ospf neighbor *, Description: This command resets the OSPF adjacency, including the neighbor relationships and the OSPF database. This can help resolve OSPF adjacency loss issues.

- Step: Clear ISIS LSP database, Command: clear isis database *, Description: This command clears the ISIS LSP database, including the LSPs and their sequence numbers. This can help resolve ISIS LSP flooding issues.